

FORGE

PURE SINE WAVE INVERTER

230063B **ATHUKORALA U R**

230121D **DE MEL D J**

230186E **FERNANDO W H D**

230508V **RAHUL B**

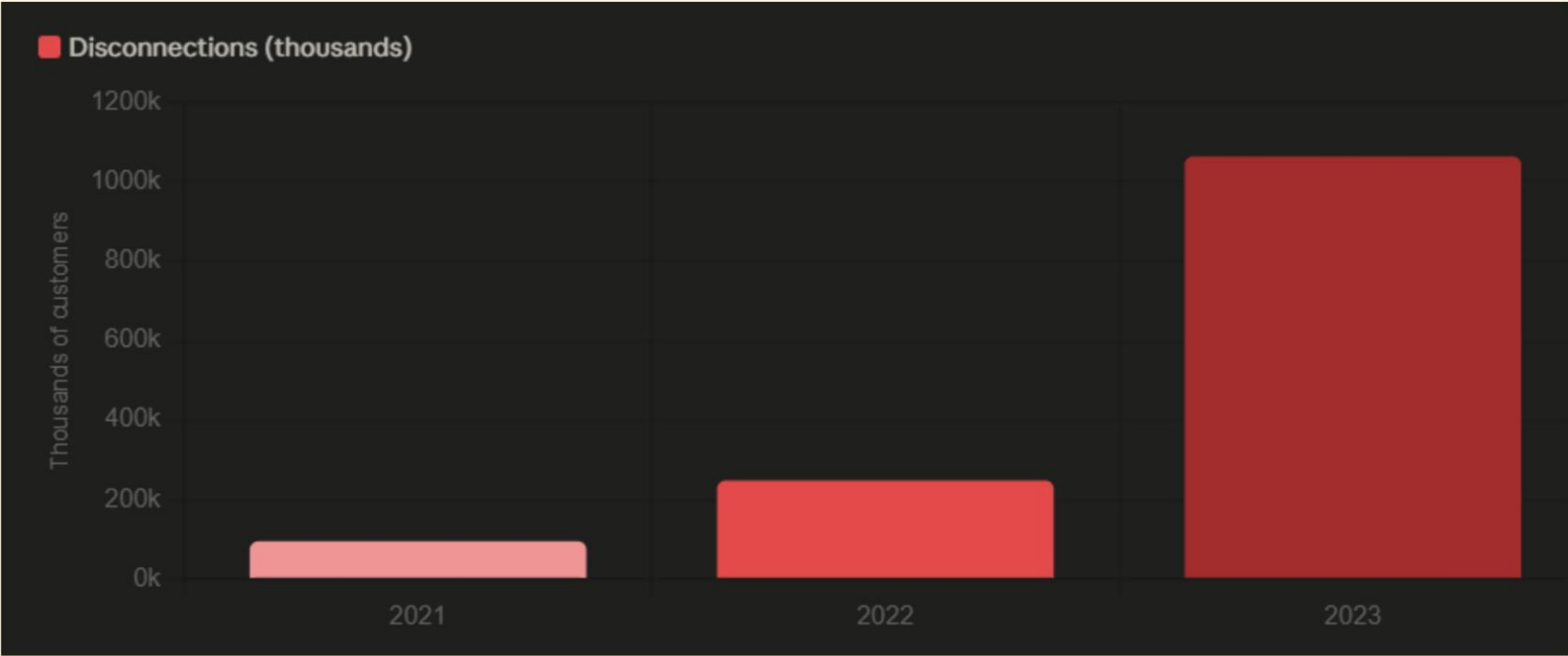
230539P **RATHEESHAN A R**



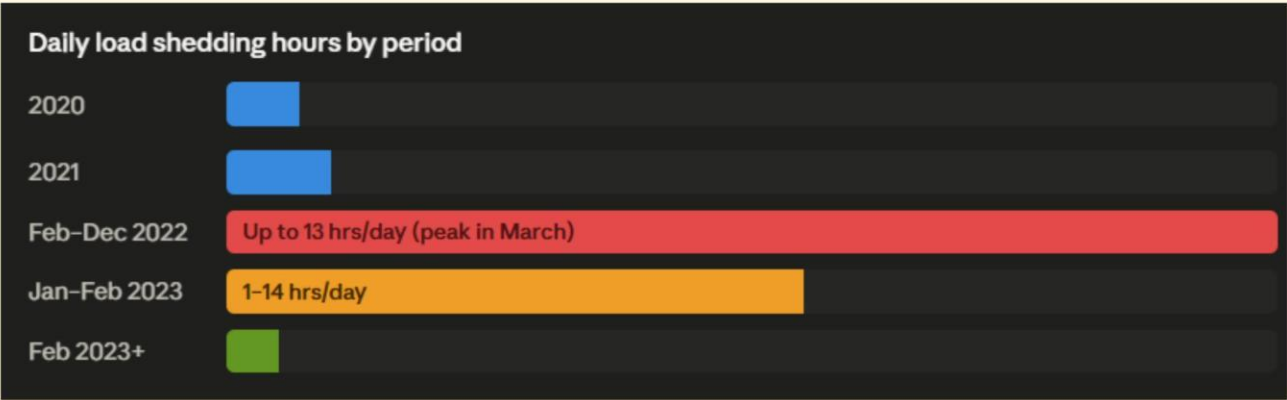
WHY THIS PROJECT?

Srilanka's power sector has experienced difficulties maintaining stability over the past 5 years, ever since the economic collapse. With tensions arising in a global scale, it is more likely that power outages will be more frequent than ever. Nowadays, people have solar panels at home which can be easily converted into AC power to use during these power outages. Our product coupled with a solar battery can produce AC current, which can be used in emergency times.

STATISTICS



Over 1 Million Sri Lankan Households Lost Power Access in 2023



Scheduled Power Cut Hours Per Day in Sri Lanka

https://www.business-standard.com/article/international/sri-lanka-returns-to-non-stop-power-supply-after-a-year-at-higher-cost-123021600679_1.html

<https://www.gazette.lk/2026/03/power-cuts-sri-lanka-ceylon-electricity-board-power-cut-schedule.html>

TECHNICAL FEASIBILITY



Component availability

SG3525, EG8010, IR2110, MOSFETs and the ferrite core transformer are all sourced from established local/regional suppliers within budget — no custom or hard-to-source parts.



Power target is achievable

300W continuous output is well within the rated capability of the chosen MOSFETs and transformer design, based on datasheet ratings and the push-pull topology selected.



Efficiency target is realistic

85–95% efficiency is consistent with ferrite-core high-frequency conversion (vs. lossier iron-core alternatives), matching results reported for comparable SG3525/EG8010-based designs.



Thermal budget is manageable

Estimated 30–45W heat dissipation at full load is handled by the combined active (fan) + passive (heatsink) cooling built into the enclosure design.



Team skillset is in place

The group has hands-on experience across PCB design, power electronics and CAD enclosure design from prior coursework, covering every stage of this build.

KEY SPECS:

Input: 12V DC

Output: 230V AC, 50Hz

Efficiency : 85% - 95%

Rated Power: 300W

Cooling: Active (Fan) + Passive (Heatsink)

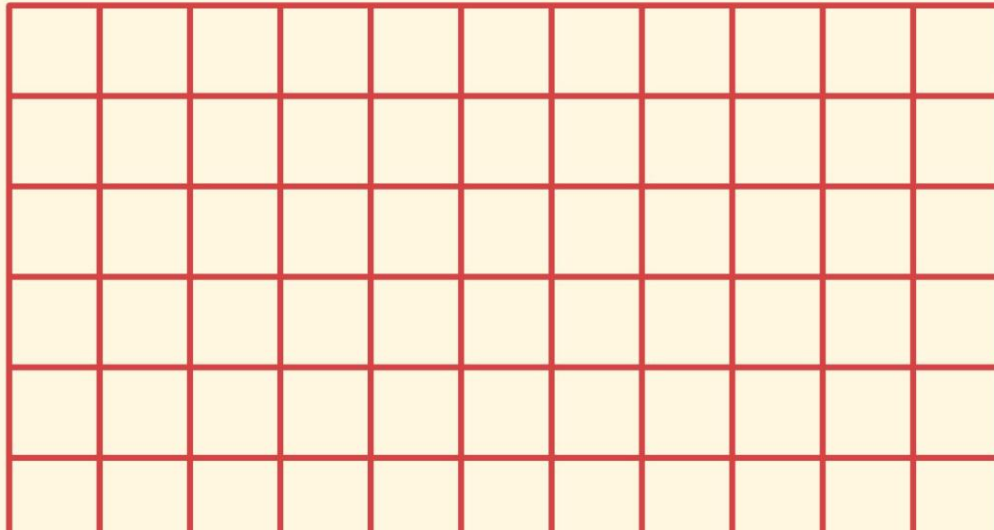
Display: LCD showing real-time voltage, current and power

Protection: Over Current, Over Voltage, Reverse Polarity.

Heat Dissipation : 30W - 45W

COMPONENTS:

IR2110 High-Low side driver: Driving the H-Bridge
Can sink and source ~2A.



CHOOSING THE PUSH-PULL CONVERTER TOPOLOGY

THERE ARE 2 METHODS TO INVERT 24V DC TO 230V AC.

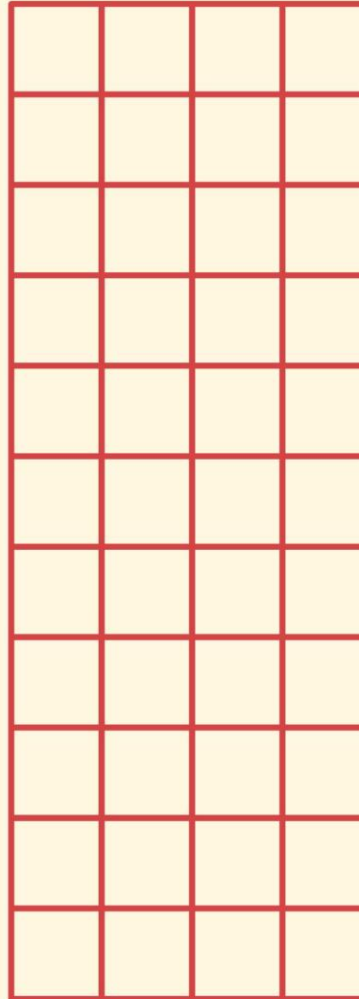
1. 12V DC – 12V AC – 230V AC

INEFFICIENT AS WE HAVE TO USE A IRON CORE TRANSFORMER TO STEP UP THE VOLTAGE.

2. 12V DC – 350V DC – 230V AC

THE POWER LOSS IN IRON CORE TRANSFORMERS IS MORE THAN FERRITE CORE TRANSFORMERS (USED IN DC-DC CONVERTERS).

WE DECIDED TO GO WITH THE 2ND METHOD BECAUSE OF IT'S HIGHER EFFICIENCY SMALLER SPACE UTILIZATION AND SIMPLER DESIGN REQUIREMENTS.



FINAL PRODUCT DETAILS



INPUT

12V DC

OUTPUT

230V AC, 50Hz pure sine wave

RATED POWER

300W continuous

EFFICIENCY

85% – 95%

BATTERY LIFE

N/A — runs off an external 12V source, not a built-in battery.
Example: a 60Ah source at 150W load gives ~4 hrs runtime.

PROTECTION

Over-current, over-voltage, reverse polarity

PCB LAYOUT & STACK JUSTIFICATION

Power Supply + H-Bridge Board

2-layer, 2oz copper

- Heavier copper and wide traces on the H-bridge / push-pull paths to carry high current with minimal I^2R loss.
- A dedicated ground plane improves thermal spreading and reduces switching noise.
- 350V DC and mains-side traces kept with generous clearance from low-voltage control lines for safety.

Controller (Brain) Board

2-layer, 1oz copper

- Signal-level currents only (EG8010 SPWM generation, feedback, LCD) — a lighter 1oz layer keeps cost down.
- Two layers give enough routing room for the SPWM/feedback nets without needing a 4-layer stack.
- Kept physically separate from the power board per the hierarchical design, isolating noisy switching from sensitive control signals.

FORGE COMPETITORS

	Tier 1 — Pure Sine Wave	Tier 2 — Hybrid Solar	Tier 3 — Generators
Input	12V DC battery	Solar + Grid	Petrol / Diesel
Brands	Unitec, SAKO, Growatt, Sorotec	Goodwe, Huawei, Victron, Growatt Hybrid	Yamaha EF, Honda EU, Local brands
Advantages	Clean output, portable, locally available	Solar integration, smart monitoring, long warranty	High power, no battery needed, widely available
Disadvantages	Import markup, limited local service, 110V/60Hz mismatch	Overkill (3–10kW), very expensive, fixed install only	Fuel cost, loud, heavy, exhaust emissions
Price Range	LKR 25,000–40,000	LKR 200,000+	LKR 40,000–400,000

OUR ADVANTAGES

- Reasonable Prices
 - 230V/50HZ native requirements
 - Local Servicing and fast support
- Optimized for Srilankan Conditions
 - Poratble

MARKET ANALYSIS

Main Customers:

- House owners : During power Outages.
- Vehicle owners : Run AC appliances with car battery
- Small Businesses : For Uninterrupted Operation
- Hikers and Campers : Portable off grid powers

Revenue Streams:

- Unit sales ,Optional extended warenty ,After sales servicing

Distribution Channels:

- Eletronic shops.
- Partnering with Solar Battery manufacturers.
- Online marketplace.

COSTING AND SELLING PRICE

MANUFACTURING PRICE (RS)

Per Unit

PCB Fabrication = RS. 3,000

Enclosure = RS. 3,000

Components = Rs. 8,000

Manufacturing cost = Rs.500

Other Expences = RS.1,500

ESTIMATED SELLING PRICE (RS)

RS.25,000

Profit margin :

Rs.9,000

MAKING IT MARKET READY

- Getting Proper certification and and safety measures.
- Reliability Testing, Regulatory Compliance.
- Improving user experience following a Human Centered Design.
LEDs, Displays, labeling to ensure easy use.
- Environmental requirements: Durable to be used under Sri Lankan temperature conditions.
- Proper Packaging and Documentation (User Manual)

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PROJECT BUDGET — BILL OF QUANTITIES

Approx. breakdown of the Rs. 8,000 components budget (per unit)

Item	Qty	Unit Cost (Rs.)	Total (Rs.)
EG8010 SPWM controller	1	1200	1200
SG3525 PWM controller	1	250	250
IR2110 high/low-side driver	2	300	600
Power MOSFETs (H-bridge + push-pull)	6	350	2,100
Ferrite core transformer (custom wound)	1	1,200	1,200
LM2678SX-5.0 buck regulator	1	350	350
TPS55289 buck-boost converter	1	500	500
Bridge rectifier + filter capacitors	1 set	850	850
LCD display module	1	650	650
Connectors, resistors, misc. passives	1 set	6,000	6,000
TOTAL			13,700

FUTURE IMPROVEMENTS

- Bluetooth/Wi-Fi monitoring app for real-time voltage, current and power tracking
- Higher power variants (500W–1000W) for wider appliance compatibility
- Integrated MPPT solar charge controller for direct solar-to-AC operation
- Smaller, lighter enclosure through a higher switching frequency (reduced magnetics size)
- Low-voltage battery cutoff with a clearer charge-level indicator
- Third-party safety certification (e.g. CE/local standards) ahead of volume production

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TASK ALLOCATION

ID Number	Task Allocation
230063B	Schematics, Testing and Assembly
230121D	Power Stage simulation and schematics, PCB design, Assembly and Testing
230186E	Enclosure design, Assembly and Testing
230508V	Controller Schematics, PCB design and routing, Component Selection, Assembly and Testing
230539P	Controller Schematics and H-bridge & Filter simulation, PCB design and routing, Assembly and Testing

CHALLENGES



- Difficulty in Winding the Transformer for our requirement.
- Difficulty in Maintaining Huge stepups like 12V to 400V under load.
- Dangers Of circuit testing with High Voltages



THANK YOU

FORGE — 12V to 230V Pure Sine Wave Inverter